

Comp 231 ceaser cipher method

Newton Kiogora

S17/03156/23

This program demonstrates the Caesar Cipher encryption technique with a shift key of 17. It allows users to input a message, which is then encrypted by shifting each letter forward by 17 positions in the alphabet. The encrypted message is then decrypted by reversing the shift, restoring the original text. The implementation uses HTML for structure, CSS for basic styling, and JavaScript for encryption and decryption logic. Non-alphabet characters such as spaces and commas remain unchanged for ensuring readability and easy understanding to the reciever

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Caesar Cipher</title>
```

```
  <style>
```

```
    body { font-family: Arial, sans-serif; text-align: center; margin: 50px; }
```

```
    textarea { width: 300px; height: 100px; }
```

```
    button { margin: 10px; padding: 10px; }
```

```
  </style>
```

```
</head>
```

<body>

<h2>Caesar Cipher method using Key as 17</h2>

<textarea id="message" placeholder="Enter your message..."></textarea>

<button onclick="processText()">Encrypt & Decrypt</button>

<h3>Encrypted Message:</h3>

<p id="encrypted"></p>

<h3>Decrypted Message:</h3>

<p id="decrypted"></p>

<script>

```
function caesarCipher(text, key, encrypt = true) {
```

```
    let result = "";
```

```
    let shift = encrypt ? key : -key
```

```
    for (let char of text) {
```

```
        if (char.match(/[a-zA-Z]/)) {
```

```
            let base = char === char.toUpperCase() ? 65 : 97;
```

```
            let newChar = String.fromCharCode(((char.charCodeAt(0) - base + shift + 26) % 26)
+ base);
```

```
            result += newChar;
```

```
    } else {

        result += char;

    }

}

return result;

}

function processText() {

    let message = document.getElementById("message").value;

    let encryptedMessage = caesarCipher(message, 17, true);

    let decryptedMessage = caesarCipher(encryptedMessage, 17, false);

    document.getElementById("encrypted").textContent = encryptedMessage;

    document.getElementById("decrypted").textContent = decryptedMessage;

}

</script>

</body>

</html>
```

Results

Caesar Cipher method using Key as 17

```
i promise to attend all lectures so that  
i can pass my exams
```

Encrypt & Decrypt

Encrypted Message:

z gifdzjv kf rkkveu rcc cvtklivj jf kyrk z tre grij dp vordj

Decrypted Message:

i promise to attend all lectures so that i can pass my exams